Homework Assignment 2

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1 problem 1: Spacetravel

Taking $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$, we know:

$$\overline{\Delta t} = \frac{1}{\gamma}\Delta t$$
$$\overline{\Delta x} = \gamma \Delta x$$

with the overline variables the measured in the reference frame of the spaceship, i.e. how the crew measures these quantities, and the not-overline variables measured from earth, or a frame of reference where both planets are stationary to one another. We know: $\overline{\Delta t} = 8y$ and $\Delta x = 8ly$. Then:

$$v = \frac{\Delta x}{\Delta t} = \frac{1}{\gamma}\frac{\Delta x}{\overline{\Delta t}} \tag{1}$$

where $\frac{\Delta x}{\overline{\Delta t}} = c$, thus:

$$\begin{aligned} v &= \sqrt{1-\frac{v^2}{c^2}}c \\ \Leftrightarrow \frac{v^2}{c^2} &= 1 - \frac{v^2}{c^2} \\ \Leftrightarrow v &= \sqrt{\frac{1}{2}}c = 2.12 \cdot 10^8 m/s \end{aligned}$$

This is the speed measured by the observer on earth, however the crew will measure:

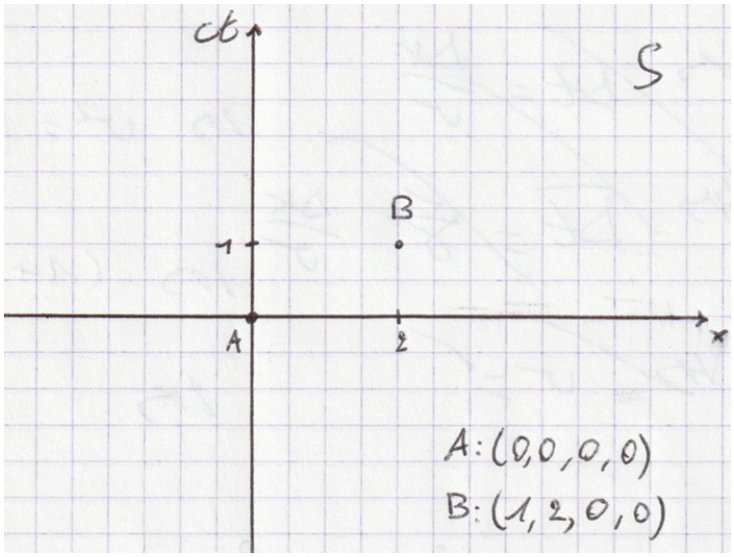
$$\begin{aligned} \overline{v} &= \frac{\overline{\Delta x}}{\overline{\Delta t}} \\ &= \gamma^2 \frac{\Delta x}{\Delta t} \\ &= \gamma^2 v \\ &= (1-\frac{1}{2})\sqrt{\frac{1}{2}}c \\ &= \frac{1}{2}\sqrt{\frac{1}{2}}c \end{aligned}$$

Thus the crew measures half the speed the observer on earth measures. This is due to the combined effects of time-dilation and length-contraction.

2 problem 2: Space-time diagrams

2.1 diagram

For s space-time diagram with position on the x-axis and ct on the y-axis, we have:



Figuur 1: Spacetime diagram showing points A and B

2.2 moving frame: simultanious events

The Lorentz-transformations say, with $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$ and $\beta = \frac{v}{c}$:

$$\begin{aligned}\overline{x^0} &= \gamma(x^0 - \beta x^1)\\ \overline{x^1} &= \gamma(x^1 - \beta x^0)\\ \overline{x^2} &= x^2\\ \overline{x^3} &= x^3\end{aligned}$$

we want $\overline{x^0_A} = \overline{x^0_B}$, thus:

$$\begin{aligned}x^0_A - \beta x^1_A &= x^0_B - \beta x^1_B\\ \Leftrightarrow 0 &= x^0_B - \beta x^1_B\\ \Leftrightarrow \beta &= \frac{x^0_B}{x^1_B}\\ &= \frac{1}{2}\\ \Leftrightarrow v &= \frac{1}{2}c\end{aligned}$$

2.3 moving frame: ct=1 earlier

we want $\overline{x^0_B} = \overline{x^0_A} - ct$, thus:

$$\begin{aligned}\gamma(x^0_B - \beta x^1_B) &= -1\\ \Leftrightarrow \beta &= \frac{x^0_B + 1/\gamma}{x^1_B}\\ \Leftrightarrow \frac{v}{c} &= \frac{1}{2} + \frac{1}{2}\sqrt{1-\frac{v^2}{c^2}}\\ \Leftrightarrow \frac{v^2}{c^2} - \frac{v}{c} + \frac{1}{4} &= \frac{1}{4}(1-\frac{v^2}{c^2})\\ \Leftrightarrow \frac{5v^2}{4c^2} - \frac{v}{c} &= 0\\ \Leftrightarrow \frac{v}{c} = 0 \wedge \frac{v}{c} &= \frac{4}{5}\end{aligned}$$

We know that v cant be zero, since then $\overline{x^0_B} \neq \overline{x^0_A} - ct$, thus we have $v = \frac{4}{5}c$.

2.4 moving frame: at the same position

We want: $\overline{x_A^1} = \overline{x_B^1}$, thus:

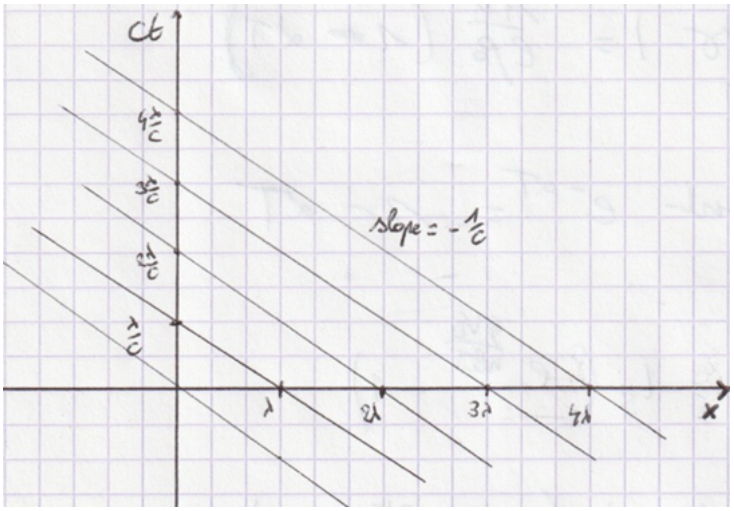
$$\begin{aligned} x_A^1 - \beta x_A^0 &= x_B^1 - \beta x_B^0 \\ \Leftrightarrow \beta &= \frac{x_B^1}{x_B^0} \\ &= 2 \\ \Leftrightarrow v &= 2c \end{aligned}$$

which isn't possible.

3 problem 3: Doppler effect

3.1 Diagram

from the equation $x = -ct + n\lambda$ we find that the slope is $-1/c$, the loci intersect the x-axis at $n\lambda$ and the ct-axis at $n\lambda/c$.



Figuur 2: Spacetime diagram showing the loci for n=0,1,2,3,4. All of these loci are parallel since they have the same slope -1/c.

3.2 Doppler equation

Consider the source moving away from the observer. However, we first start bij looking at the reference frame of the source. This means that the observer is moving away from the source at speed v. When the observer is a distance l away from the source it sends out a lightpulse. According to the source's reference frame this pulse meets the observer after a time $t_1 = \frac{l+vt_1}{c}$ or thus $t_1 = \frac{l}{c-v}$. Let T be the period of the lightpulses. The second pulse meets the observer at a time $t_2 = \frac{cT+l}{c-v}$. So, according to the source, the observer measures a period $T_{obs} = t_2 - t_1 = \frac{T}{1-v/c}$. However, call $\overline{T}$ the period according to the observer, then due to relativity:

$$\Delta t = T_{obs} = \gamma \overline{\Delta t} = \gamma \overline{T} \tag{2}$$

Which gives us:

$$\overline{T} = \frac{\sqrt{1 - v^2/c^2}}{1 - v/c} T = \sqrt{\frac{1 + v/c}{1 - v/c}} T \tag{3}$$

Now we need to remember in which reference frame we are. λ is the wavelength for the observer, who observes a period $\overline{T}$, and λ' is the wavelength according to the source, which observes a period T. Since $T = \lambda'/c$, we find indeed:

$$\lambda' = \sqrt{\frac{1 - v/c}{1 + v/c}} \lambda \tag{4}$$

4 problem 4: Particle collider

In the reference frame of the collider the particles have energies:

$$\begin{aligned} E_{p^+} &= \frac{mc^2}{\sqrt{1-v^2/c^2}} = 2m_pc^2 \\ E_{p^-} &= 2m_pc^2 \end{aligned}$$

thus the total energy, before the collision is, with $m_p = 938MeV/c^2$:

$$E_b = 4m_pc^2 \tag{5}$$

After the collision the energy is, for n particles with $m_e = 134MeV/c^2$:

$$E_f = \sum_{i=1}^{n} \frac{mc^2}{\sqrt{1-v_i^2/c^2}} \tag{6}$$

with v_i the velocity of one particle in the x-direction. However, in order to get the most amount of particles, we need $v_i = 0$ for each particle, since then the function $\frac{1}{\sqrt{1-v_i^2/c^2}}$ is at a minimum. Due to conservation of energy we have:

$$\begin{aligned} E_b &= E_f \\ 4m_pc^2 &= \sum_{i=1}^{n} \frac{mc^2}{\sqrt{1-v_i^2/c^2}} \\ nm_ec^2 &= 4m_pc^2 \\ n &= \frac{4m_p}{m_e} \\ &= 28 \end{aligned}$$

thus, we can get a maximum of 28 particles.

5 problem 5: Lorentz boost

Using equations 12.111 in the handbook, we can transform these by substituting $x \to z$ and $z \to x$, thus via $\overline{\mathbf{E}} = \gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ and $\mathbf{B} = -\frac{1}{c^2}(\mathbf{v} \times \overline{\mathbf{E}})$ we find:

$$\begin{aligned} \overline{E}_x = \gamma(E_x - vB_y), \overline{E}_y = \gamma(E_y + vB_x), \overline{E}_z = E_z \\ \overline{B}_x = \gamma(B_x + \frac{v}{c^2}E_y), \overline{B}_y = \gamma(B_y - \frac{v}{c^2}E_x), \overline{B}_z = B_z \end{aligned}$$

6 problem 6: Field tensor

6.1 covariant field tensor

$$F_{\mu\nu} = \begin{bmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & B_z & -B_y \\ E_y/c & -B_z & 0 & B_x \\ E_z/c & B_y & -B_x & 0 \end{bmatrix} \tag{7}$$

6.2 Levi-Civita

You need three permutations from 0123 to get to 1230:

$$0123 \to 1023 \to 1203 \to 1230 \tag{8}$$

since there is an uneven amount of permutations, we have $\varepsilon^{1230} = -\varepsilon^{0123} = -1$

6.3 dual field tensor

If there are repetitions of indexes, then $\varepsilon^{\mu\nu\lambda\sigma} = 0$. This means that we can discard all $F_{\lambda\sigma}$ for which either λ or σ are 0 or 1. This means only the components of F_{23} and F_{32} remain, giving us:

$$\tilde{F}^{01} = \frac{1}{2}B_x + \frac{1}{2}B_x = B_x \tag{9}$$

For $\tilde{F}^{02}$ only F_{13} and F_{31} remain, giving us:

$$\tilde{F}^{02} = \frac{1}{2}B_y + \frac{1}{2}B_y = B_y \tag{10}$$

For $\tilde{F}^{12}$ only F_{03} and F_{30} remain, giving us:

$$\tilde{F}^{12} = \frac{1}{2}E_z/c + \frac{1}{2}E_z/c = E_z/c \tag{11}$$

For $\tilde{F}^{13}$ only F_{02} and F_{20} remain, giving us:

$$\tilde{F}^{13} = -\frac{1}{2}E_y/c - \frac{1}{2}E_y/c = -E_y/c \tag{12}$$

6.4 expression for $F^{\mu\nu}$

Look at $\tilde{F}^{\mu\nu}$ compared to $F_{\lambda\sigma}$. The net effect of the $\frac{1}{2}\varepsilon^{\mu\nu\lambda\sigma}$ is that **E/c** $\rightarrow$ **-B** and **B** $\rightarrow$ **-E/c**, like discribed on page 568 of the textbook. Now we have $\tilde{F}_{\lambda\sigma}$:

$$\tilde{F}_{\lambda\sigma} = \begin{bmatrix} 0 & -B_x & -B_y & -B_z \\ B_x & 0 & -E_z/c & E_y/c \\ B_y & E_x/c & 0 & -E_x/c \\ B_z & -E_x/c & E_x/c & 0 \end{bmatrix} \tag{13}$$

compared to:

$$F^{\mu\nu} = \begin{bmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{bmatrix} \tag{14}$$

we see that we exactly need to substitute **E/c** $\rightarrow$ **-B** and **B** $\rightarrow$ **-E/c**. Thus we find:

$$F^{\mu\nu} = \frac{1}{2}\varepsilon^{\mu\nu\lambda\sigma}\tilde{F}_{\lambda\sigma} \tag{15}$$